Go2 SDK Development Guide

Search article titles

About Go2
Remote Controller
SDK Concepts 3
Application Development 7
Software Interface Services 19
Motion Services Interface V2.0
DDS Services Interface
Avoidance Services Interface
VuiClient Interface
RobotStateClient Interface
Motion Switcher Service Interface
Sports Services Interface
AI motion Services interface
Basic Services Interface
LiDAR Services Interface
Multimedia Services Interface
UWB Services Interface
Fault Services Interface
SLAM and Navigation Services Interfa
Aruco Recharge Service Interface
Odometer Services Interface
D1 Mechanical Arm Services Interface
ROS2 Services Interface
Python Services Interface
Case reference 8
Debugging Specification
FAQ



# Avoidance Services Interface

Last Updated On : 2025-11-21 11:19:40

This page contains content

ObstaclesAvoidClient Class
Interface Error Code List:
Class and Interface Descripti...

## ObstaclesAvoidClient Class

ObstaclesAvoidClient is the client provided by the Go2 obstacle avoidance service. Through ObstaclesAvoidClient, you can use the RPC method to enable/disable and get the status of the Go2 obstacle avoidance function.

### Interface Error Code List:

This Client does not have a specific error list. For the general list, please refer to: List of some general error codes

### Class and Interface Description:

| Class Name | Constructor and Destructor |
|---|---|
| ObstaclesAvoidClient | explicit ObstaclesAvoidClient(); ~ObstaclesAvoidClient(); |

| Function Name | SwitchSet |
|---|---|
| Function Prototype | int32_t SwitchSet(bool enable) |
| Function Summary | Sets the obstacle avoidance function to be enabled or disabled. |
| Parameters | **enable**: Set to true to enable, false to disable. |
| Return Value | Returns 0 on success, otherwise returns the relevant error code. |
| Notes | |

| Function Name | SwitchGet |
|---|---|
| Function Prototype | int32_t SwitchGet(bool& enable); |
| Function Summary | Gets whether the obstacle avoidance function is enabled. |
| Parameters | **enable**: Return value, true for enabled, false for disabled. |
| Return Value | Returns 0 on success, otherwise returns the relevant error code. |
| Notes | |

| Function Name | UseRemoteCommandFromApi |
|---|---|
| Function Prototype | int32_t UseRemoteCommandFromApi(bool isRemoteCommandsFromApi) |
| Function Summary | Takes over the speed command control from the remote controller. |
| Parameters | **isRemoteCommandsFromApi**: Set to true to enable takeover, false to disable takeover. |
| Return Value | Returns 0 on success, otherwise returns the relevant error code. |
| Notes | `isRemoteCommandsFromApi` must be set to true to control dog obstacle avoidance via API. |

| Function Name | Move |
|---|---|
| Function Prototype | int32_t Move(float x, float y, float yaw); |
| Function Summary | Movement control in obstacle avoidance mode. |
| Parameters | **x**: Body x-direction velocity command, limited to -1.5~+1.5(m/s); **y**: Body y-direction velocity command, limited to -1~+1(m/s); **yaw**: Body yaw angular velocity command, limited to -1.57~+1.57(rad/s). |
| Return Value | Returns 0 on success, otherwise returns the relevant error code. |
| Notes | `isRemoteCommandsFromApi` must be set to true to control dog obstacle avoidance via the Move function. |

| Function Name | MoveToIncrementPosition |
|---|---|
| Function Prototype | int32_t MoveToIncrementPosition(float x, float y, float yaw); |
| Function Summary | Movement control in obstacle avoidance mode. |
| Parameters | **x**: Incremental position in the body's x-direction, limited to -2.0~+2.0(m); **y**: Incremental position in the body's y-direction, limited to -2.0~+2.0(m); **yaw**: Incremental yaw angle, limited to -1.57~+1.57(rad). |
| Return Value | Returns 0 on success, otherwise returns the relevant error code. |
| Notes | `isRemoteCommandsFromApi` must be set to true to control dog obstacle avoidance via the MoveToIncrementPosition function. |

| Function Name | MoveToAbsolutePosition |
|---|---|
| Function Prototype | int32_t MoveToAbsolutePosition(float x, float y, float yaw); |
| Function Summary | Movement control in obstacle avoidance mode. |
| Parameters | **x**: Absolute position in the world coordinate system x-direction; **y**: Absolute position in the world coordinate system y-direction; **yaw**: Yaw angle in the world coordinate system. |
| Return Value | Returns 0 on success, otherwise returns the relevant error code. |
| Notes | `isRemoteCommandsFromApi` must be set to true to control dog obstacle avoidance via the MoveToAbsolutePosition function. |

Example:

```cpp
#include <unitree/robot/go2/obstacles_avoid/obstacles_avoid_client.hpp>
```

Helpful
Feedback

```
using namespace unitree::common;
using namespace unitree::robot;
using namespace unitree::robot::go2;

int main()
{
    unitree::robot::ChannelFactory::Instance()->Init(0, "enx00e04c36141b");//enx00e04c36141b is the network port
    unitree::robot::go2::ObstaclesAvoidClient sc;
    sc.SetTimeout(5.0f);
    sc.Init();
    sc.SwitchSet(true);//Enable obstacle avoidance
    usleep(1000000);
    sc.UseRemoteCommandFromApi(true);//Take over speed control from the remote controller
    sc.Move(1.0,0.0,0.0);//Move along the body's x-direction at 1m/s, automatically avoiding obstacles when encou
    int tem = 0;
    while (true)
    {
      usleep(1000000);
      if(tem<=5)
      {
        tem++;
      }
      else
      {
        sc.UseRemoteCommandFromApi(false);//Disable after 5 seconds
        sc.SwitchSet(false);//Disable after 5 seconds
      }
    }
    return 0;
}
```

The PDF has been downloaded. Rate us: ☆☆☆☆☆